Kamilla Kaarsholm, Peter Kjeldsen and Henrik Andersen

Grand exercise 2025 – Inappropriate disposal of a pesticide in a gravel pit

Introduction

In a small town on Southern Funen, a farmer discovers a 100 kg bag of "Præfix" standing at the back of his farm shed. Præfix is an herbicide that contains the active compound dichlobenil (10% by weight). Dichlobenil has been banned in Denmark since 1997 and the farmer does not want to use it on his fields, as his neighbors are very environmentally conscious, and he wants to stay on good terms with them. However, he still wants to get rid of the bag. As he considers it to be a small amount, he drives to a local closed gravel pit, which today is known as *"Lake Gravel Pit"* with recreational surroundings used by locals. The farmer tips the contents of the bag into the lake and hurries away. Fortunately, an engineer from the local municipality was walking her dog nearby at the time and witnessed the incident.

The local authorities now need to evaluate several aspects of this problem, and so they have asked DTU Sustain for help in an expert evaluation of the situation, and guidance. Your group has taken on the task.

Reporting

The deadline for reporting is tight, as the authorities might need to act fast to remediate the problem. The report has to be handed in **no later than 3rd December 2025 at 10.00 pm**, as the engineer from the municipality has to prepare her presentation and recommendation for the city council meeting the following afternoon.

Based on the skills obtained in the course "Environmental processes", the report should evaluate the three main tasks listed below. The evaluation should be supported by relevant calculations, which will be an integrated part of the report.

The report is intended for the municipal engineer, and to serve as an appendix for the engineer's recommendation on the matter to the technical committee of the municipality. Therefore, the main report cannot exceed 15 pages, and the total length of additional appendices is limited to 20 pages.

Tasks

The pollution of "*Lake Gravel Pit"* with dichlobenil (DCB) and its degradation product dichlorbenzamide (BAM), means that the authorities need a technical assessment of the following aspects:

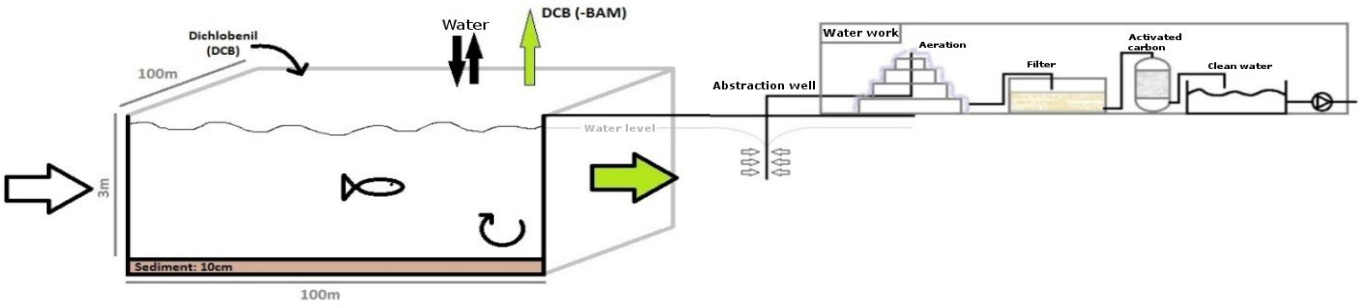
1) What will the concentration of DCB and BAM be in the lake, and would that result in direct toxicity for duckweed or fish?
2) Does the pollution cause a problem for the local waterworks? If yes, can a new treatment process (or processes) be established for pesticide treatment, or do the waterworks need to switch to another well as a water source?
3) Is the water from the other well suitable for potable water production?

N.B. More information on these tasks is detailed on the following page (see *Further information regarding tasks*).

For simplicity, it can be assumed for the calculations that the lake is quadratic with side lengths of 100 m, and depth of 3 m. The ground water intrusion in the lake is through one side (100 m · 3 m) with a Darcy velocity of 20 m/year. The groundwater is leaving the lake through the opposite side to the intrusion. The addition of water from precipitation can be neglected (i.e. precipitation is assumed to be equal to evaporation). All water from the lake is abstracted by the nearby waterworks, which produce 300,000 m^3 of potable water per year.

The physical-chemical properties and biological effects of DCB and BAM are provided in the Appendix.

N.B. In many cases, you may simplify a problem by making your own (reductive) assumptions. In such situations, it is important that these assumptions are made clear.



Further information regarding tasks:

1) Concentration of DCB and BAM in the lake

You have to describe the concentration of DCB and BAM in the lake over time (i.e. making a graph of concentration of DCB and BAM as function of time) and evaluate whether the DCB or BAM has an acute toxic effect on fish in the lake. This can be done by:

a) calculate the initial concentration of DCB. It can be assumed that the upper 10 cm of the sediment at the bottom of the lake (100m·100m) is in direct contact with the water in the lake and thus limitation due to transport can be neglected in this layer. In other words, the sediment is completely mixed with the water. The content of organic carbon in the sediment is 2%. Assume linear sorption.

b) write the mass balance (also called material balance) needed to model the different transport processes for DCB. Through these calculations, identify which processes are the main contributors to removal of DCB from the lake.

c) write the mass balance of BAM (sorption, evaporation and degradation can be neglected, but remember to justify this).

d) compare the max concentration found with the EC_{50} given in appendix. Is the water toxic for fish and duckweed?

Hint 1. Remember to include sorption to sediment when making the mass balance for DCB.

Hint 2. The main part of DCB in the lake disappears within the first 2-3 months (60-80 days). When solving the mass balance for DCB, the removal with the ground water flow can be neglected to simplify the equation. However, remember to justify this. For BAM, you can divide the solution in two: one equation for the first two-three months where DCB is disappearing, and then a second equation that describes what will happen in the following months/years.

2) At the waterworks

You should evaluate whether the pollution cause a problem for the local waterworks. Furthermore, you have to investigate the effectiveness of using activated carbon filters and the effect of changing aeration tank. This can be done by considering the following information and questions:

The polluted water from Lake Gravel Pit is transported with the groundwater, to an abstraction well belonging to the nearby waterworks. It is assumed that the concentration of BAM in groundwater originating from Lake Gravel Pit is constant, and equal to the maximum concentration in the lake (worst case). The lake water is diluted at the abstraction well.

a) Calculate the inlet concentration for the waterworks.

At the waterworks, the ground water (raw water) is treated by aeration and sand filtration.

b) Is BAM efficiently retained by sorption on the existing sand filter? (Determine the retardation factor)

The local manager of the waterworks believes that cleaning with activated carbon will be the best solution if the authorities permit it.

c) What would the dimensions be for the cylindrical container required to treat the produced volume of water? (See Appendix for information about carbon filters).
d) How often does the activated carbon need to be changed? Calculate both with and without dispersion and for both cases draw the breakthrough curve for BAM in the outlet of the carbon.

One problem which can occur when introducing activated carbon filtration, is that iron and manganese oxides can precipitate in the filter and block the pores. Thus, it is necessary to have sufficient removal of these mineral oxides before they can interact with the carbon filter. To achieve this, the waterworks has to change the existing cascade aerator into an aeration tank. Initially, the water is highly oversaturated with carbon dioxide compared to the equilibrium concentration. The aeration entails that carbon dioxide is reduced to 9 times higher than the equilibrium concentration with atmospheric air.

e) Could this result in problems with precipitation of calcite in the carbon filter? (calculate both with activity coefficient = 1, and with activity coefficient < 1).

During aeration, iron(II) is oxidized to iron(III), which then precipitates as iron hydroxide. The overall reaction (oxidation and precipitation) follows a first order reaction with a rate constant of 1.0 min^{-1}.

f) Write the overall reaction for oxidation and precipitation of iron.

For iron removal, different designs (CMFR, CMFRs in series and PFR) of the aeration and reaction tank is considered. For all designs, a total retention time at 13 min is required. In drinking water, the maximum allowed concentration of iron is 200 µg/L.

g) Calculate the outlet concentrations from and evaluate which of the reactors that meets the requirement for drinking water.

3) New abstraction well

A new ground water well is considered as a replacement of the current well as an option to avoid BAM getting into the waterworks. The raw water (before aeration and filtration) in the new well is characterized by the following parameters:

Cations	**[mg/L]**	**Anions**	**[mg/L]**
Sodium, Na^+	50	Chloride, Cl^-	200
Potassium, K^+	15	Nitrate, NO_3^-	2.6
Calcium, Ca^{2+}	210	Hydrogen carbonate, HCO_3^-	385
Magnesium, Mg^{2+}	37	Sulfate, SO_4^{2-}	205
Iron, Fe^{2+}	2		
Ammonium, NH_4^+	1		

To evaluate the new well consider the following sub-task:

a) Investigate where the water analysis is complete or if any major ion missing
b) Calculate the hardness of the water in meq/L and °dH
c) Compare with the water quality requirements for drinking water. The given quality requirements is at consumers tap which means after aeration and filtration at the waterworks. The requirements can be found at:
https://www.retsinformation.dk/eli/lta/2023/1023#idfe58b72d-9489-448d-8f93-976f3f7916a3
(Consolidation Act about water quality and inspection of waterworks units (Bekendtgørelse om vandkvalitet og tilsyn med vandforsyningsanlæg), BEK nr 1023 af 29/06/2023).

Based on your answers for the three sub-task, come with your recommendation for the new water well. Can it be used or not, would additional water treatment beside aeration and sand filtration be required or would additional information be need to before decision is made?

Reporting

The target audience of the report is the engineer from the municipality. That means you should give due consideration to the language and structure used for the report, so that an engineer could thoroughly evaluate it. A municipality engineer is skilled all-round environmental engineer, with many other assignments besides pesticide pollution and water treatment. The main report should not be technically-detailed, as it is important to provide the municipality engineer with information to make an informed decision. The report should give a good overview, be concise, and good readability in general. The principles and the main results from calculations should be explained in the outline of the report and supported by figures and tables. The detailed calculations should be located in the appendices, but referenced to in the main report (e.g. "see Appendix 1"). Remember to provide references for your information including constants, equations etc., e.g Bjerg & Kjeldsen (2004), p…., Figure…

A report to a municipality is not structured the same way that your laboratory exercise reports. It will contain sections relating to the questions asked. The report should have an attractive front page with group number, names, and student numbers. Furthermore, the report should contain a summary (approx. 1 page), an introduction with background and aim, an evaluation of the problems, a conclusion, a list of references, and the appendices.

The report should be uploaded as "Grand Exercise" under Assignments at DTU Learn, including all appendices and files needed to show detailed calculations, for example Excel files or pdf print of Maple files. Please keep in mind to have a good structure in your appendices.

Appendix – Substance properties and technical data

Pesticides

Compound	CAS	K_H (atm·m^3·mol^{-1})	Toxicity data[1,2]		
			LC_{50} (mg/L)		EC_{50} (mg/L)
			Rainbow trout	Fathead minnow	Duckweed
DCB	1194-65-6	$8.3 \cdot 10^{-6}$	3.83	3.98	0.0279
BAM	2008-58-4	$1.22 \cdot 10^{-9}$	180	496	75

LC_{50} = concentration which is lethal to 50 % of the population, EC_{50} = concentration which affects 50 % of the population, The ECOTOX database, 2004. Available on http://www.epa.gov/ecotox/, Rippen, 2003. Handbuch Umweltchemikalien CD-ROM 05/2003.

Dichlobenil, DCB
Half-life in lake: $t_{1/2}$ = 128 d (first order kinetic)
Log K_{OW} = 2.70
BAM
K_d (active carbon) = 1.15 m^3/g active carbon
Log K_{OW} = 0.77

Lake Gravel Pit

Sediment:

Bulk density, ρ_b	1600	kg/m^3

The Waterworks

Filter material in existing sand filter:

Filter depth	2.00	m
Bulk density, packet	1300	kg/m^3
Porosity	0.50	
Organic carbon	0.0011	%

Active carbon filter:

Residence time (based on empty bed) T_h	20	min
Geometry: height:diameter ratio	2	
Porosity, ϕ	0.5	
Bulk density, ρ_b	0.42	kg/L
Dispersion coefficient for pesticides, ε	10	m^2/h

Existing plant raw water analysis:

Temperature	10	°C
Oxygen	0	mg/L
pH	7.5	
TAL	4.61	meq/L
Conductivity	61	mS/m
Calcium	94	mg/L
Iron	6	mg/L